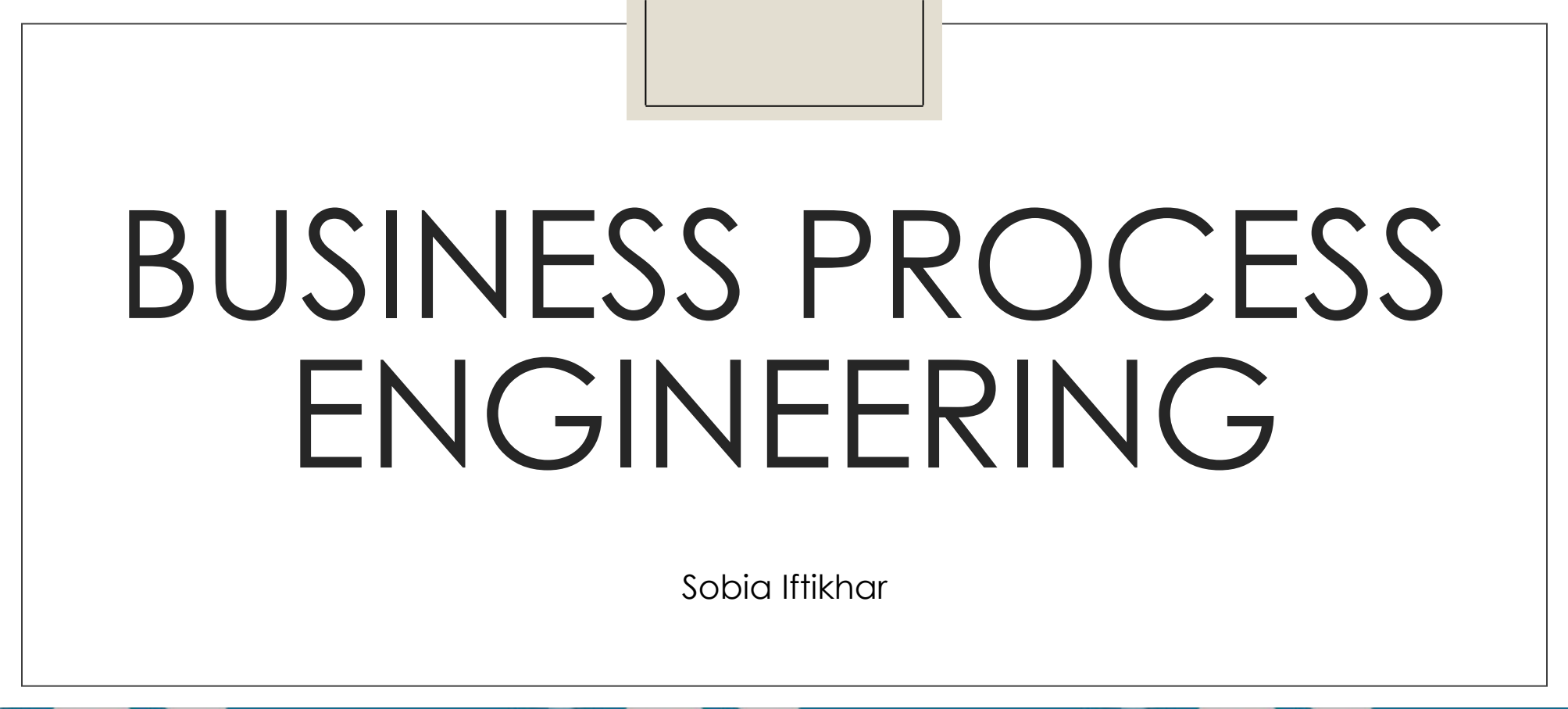
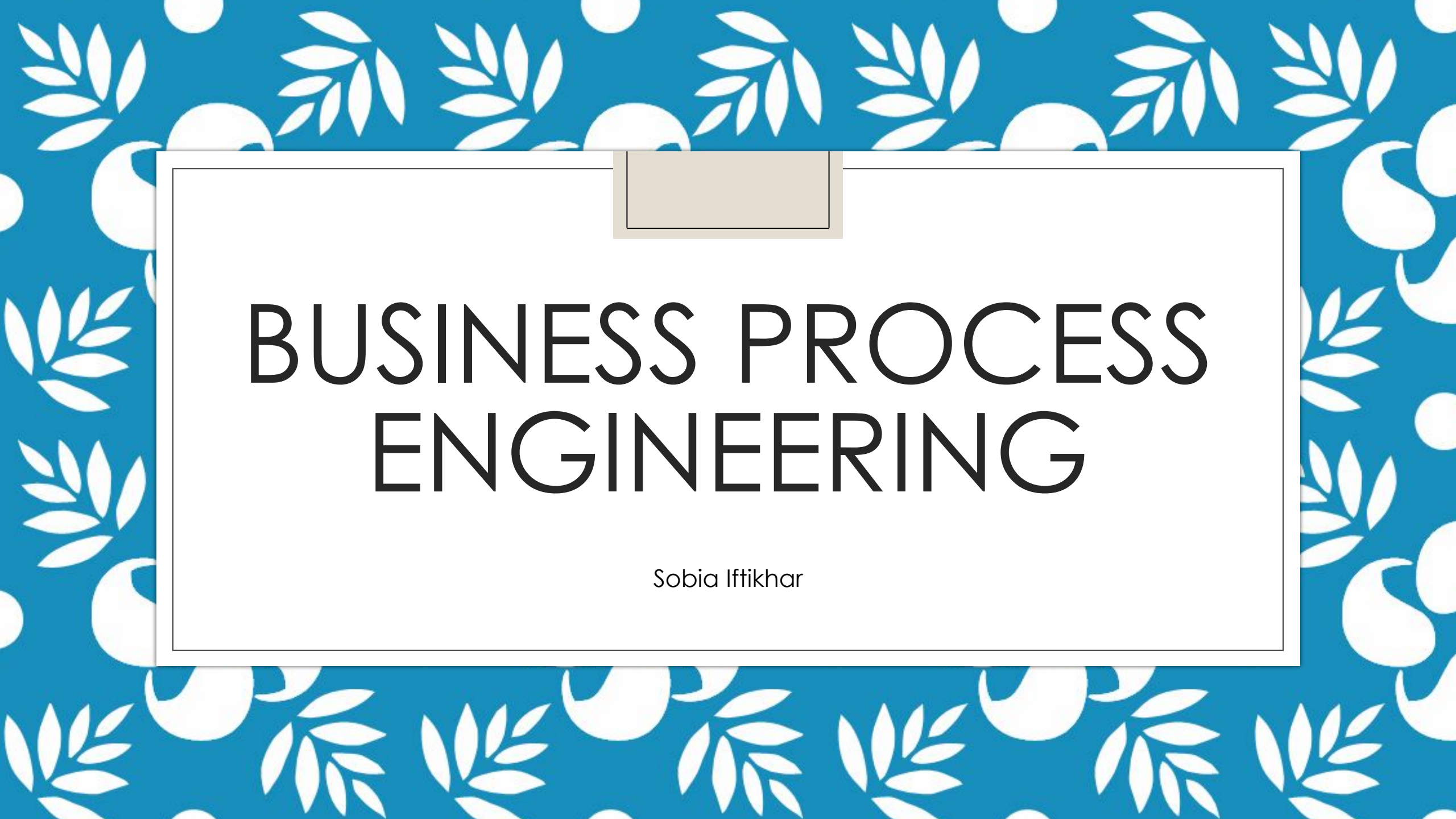




BUSINESS PROCESS ENGINEERING

Sobia Iftikhar

Content

- Business Process Improvement
- How it effects on business process
- Business Process Methodology
- (Lean Management, Six Sigma, Agile Management, Kaizen, Total Quality Management).
- Lean Management
- Six Sigma
- TQM

What is Business Process Improvement?

- Business process improvement is an exercise in which leaders use different methods to analyze their procedures.
- Business process improvement (BPI) is an approach used to identify and evaluate inefficiencies within the organization.
- It focus on following:

- redesigns existing business tasks
- improving their effectiveness
- enhances the workflows
- optimizes performance.

Business Process Improvement Steps

Step 1 – Mapping

- The first step presumes the identification of the problem

Step 2 – Analysis

- Does our company have jobs that are taking longer than required?

Step 3 – Get Buy-In

- What are the main factors influencing the inefficiencies within the organization?
- you have to display the existing problem before the senior

Step 4 – Implement

means you have to get approval for

Step 5 – Review and Refine

WHAT DOES BUSINESS PROCESS IMPROVEMENT LOOK LIKE FOR YOU?

- Reduce the time it takes to get a campaign out the door with automated communication

Marketing

- Integrate your onboarding process with your HRMS to reduce manual data transfers

HR

- Add in conditional steps to get approvals for large orders over a certain value

SALES

- Improve response time to service requests by auto-assigning tickets to specialists

IT

BPI aims to pinpoint the skills or processes that need improvement in ensuring smooth procedures, efficient workflow, and business expansion

BPI Methodologies

Challenges are common in BPI activities.

It's possible that there will be a breakdown in communication strategies across the organization, that work teams will be opposed to change, and that planned improvements that are not consistent with business objectives will be avoided.



BPI Methodologies

◦ Implementing these following methodologies can help organizations overcome previous mentioned obstacles. The common BPI methodologies include:

- ✓ Lean management
- ✓ Six sigma
- ✓ Agile management
- ✓ Kaizen
- ✓ Total quality management

Lean Management

- **Lean Management:** Lean Management is a unique approach to identify waste. It involves employees and relies on continuous improvement to maximize customer value and minimize waste.
- The main purpose of lean management is to reduce factors that waste time, effort or money. The only way to manage with it is to analyze a business process and then revise it and cut out every step that does not create value for customers

Lean Principles

- Lean management principles include:
 - Defining value from the standpoint of the end customer.
 - Identifying each step in a business process and eliminating those steps that do not create value.
 - Making the value-creating steps occur in a tight sequence.
 - A pull system is characterized by work entering the workflow only when there is a demand for it
 - Repeating the first three steps on a continuous basis until all waste has been eliminated.



The Plan, Do, Check, Act methodology

- **1. Plan**

- What is the core problem that needs solving?
- What resources do we need?
- What resources do we have?
- What is the best solution for fixing the problem with the available resources?

Also known as the Deming Cycle, the Plan-Do-Check-Act cycle is a key instrument of the lean manufacturing toolkit.

- **2. Do**

- Apply what was considered at the planning stage and put things into action

- **3. Check**

- Audit your plan's execution and see if your initial plan worked. It will become clear what the more problematic parts of the process are, which can be eliminated later down the line.

- **4. Act**

- If things didn't go so well, you'll need to begin the cycle again

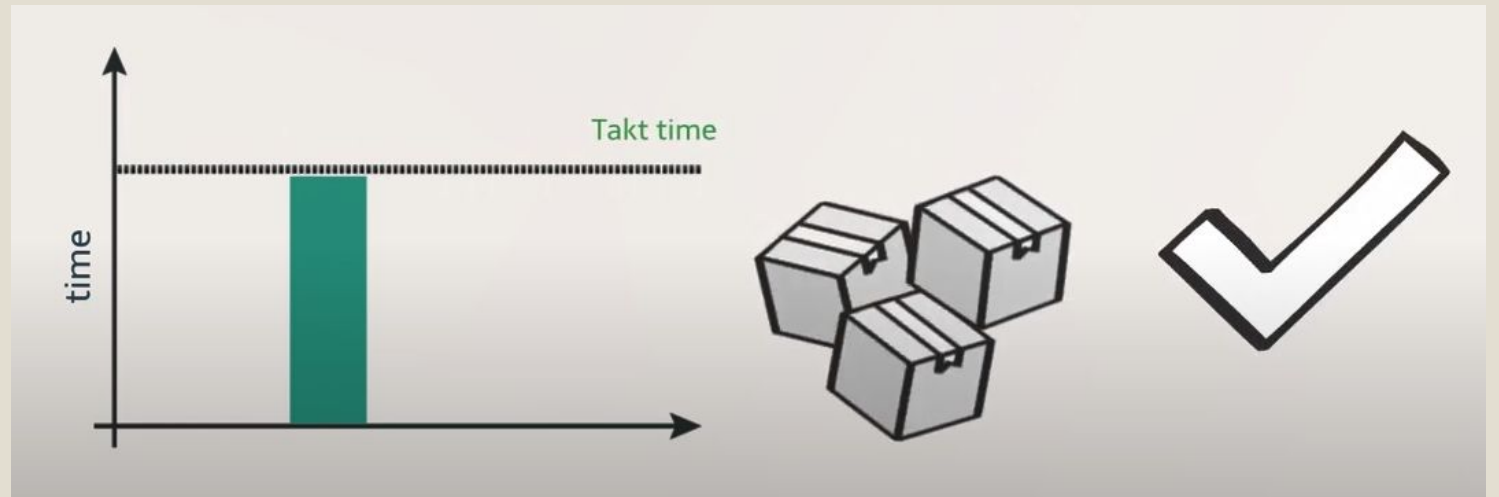
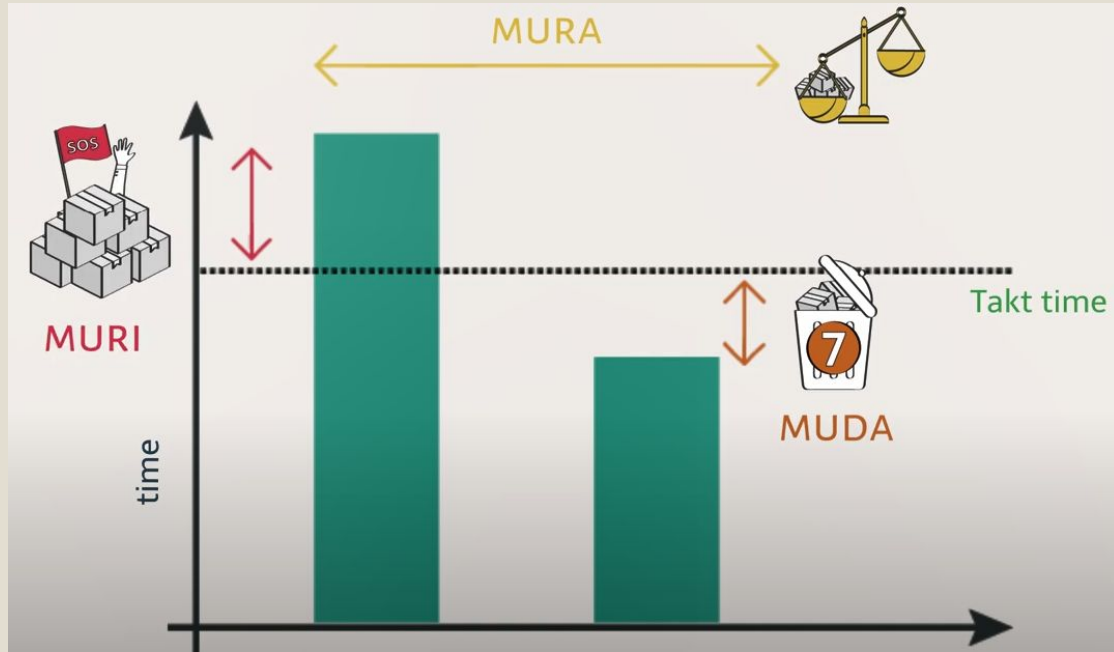
Identify the value

- Value is every thing, that your customer is willing to pay.
- Identify Waste Activities and Where Value Is Added
- **seven types of waste:** TIMWOOD
 - Transport
 - Inventory
 - Motion
 - Waiting
 - Overproduction
 - Over-processing
 - Defects

3M Waste are:

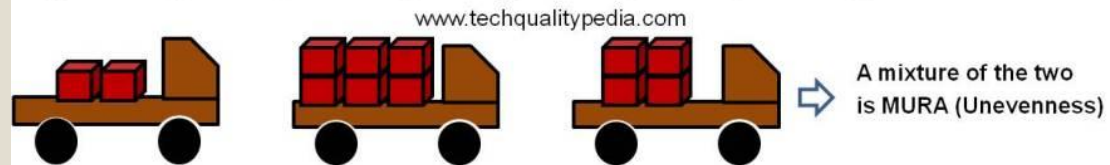
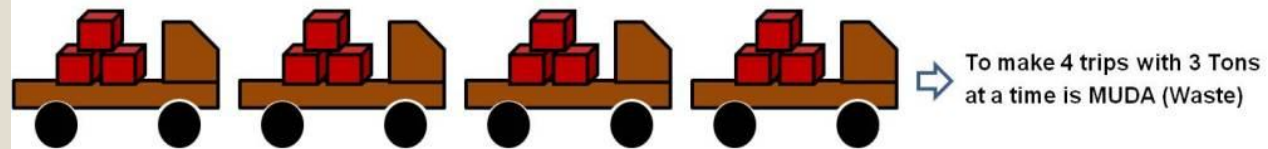
- **MUDA** means Waste
- **MURA** means Inconsistency/Unevenness/Imbalance
- **MURI** means Strain/Overburden/Fatigue

The Three waste Aspects



What is 3M Waste? MUDA, MURA, MURI

Example: Suppose you need to transport 12 Tons of material with a 4 Ton capacity Truck.



MUDA means Waste

- MUDA stands for eliminating waste or non-value-added activities from the manufacturing process. It can be anything that does not add value from the customer's point of view.
- There are **7 types of MUDA** or **7 types of Waste** identified in the lean manufacturing process which is as follow:
 - *MUDA of **Overproduction***
 - *MUDA of **Process Itself/Over Processing***
 - *MUDA of **Inventory/Stocks***
 - *MUDA of **Transport***
 - *MUDA of **Waiting***
 - *MUDA of **Motion***
 - *MUDA of **Defects***

Seven types of **Muda** or **7 types of Waste** are:

SNO	Factor	Example
01	Overproduction	occurs when items are manufactured before they are required.
02	Over Processing	occurs when more work is put into a product or service than is actually required.
03	Inventory	Inventory is important, but storing too many or too few supplies and finished goods increases costs and can take valuable space.
04	Transport	Transportation is the movement of products and supplies from one area to another.
05	Waiting	Waiting is when work or projects come to a halt or slow because machines have yet to finish producing the required goods, products have not arrived, or some other task is holding up the rest of the project.
06	Motion	Motion is the physical movement of a person or machine required to complete work.
07	Defects	Defects occur when errors and rework are created because of your processes.

Example

- **The Wastes In Lean Software Development**

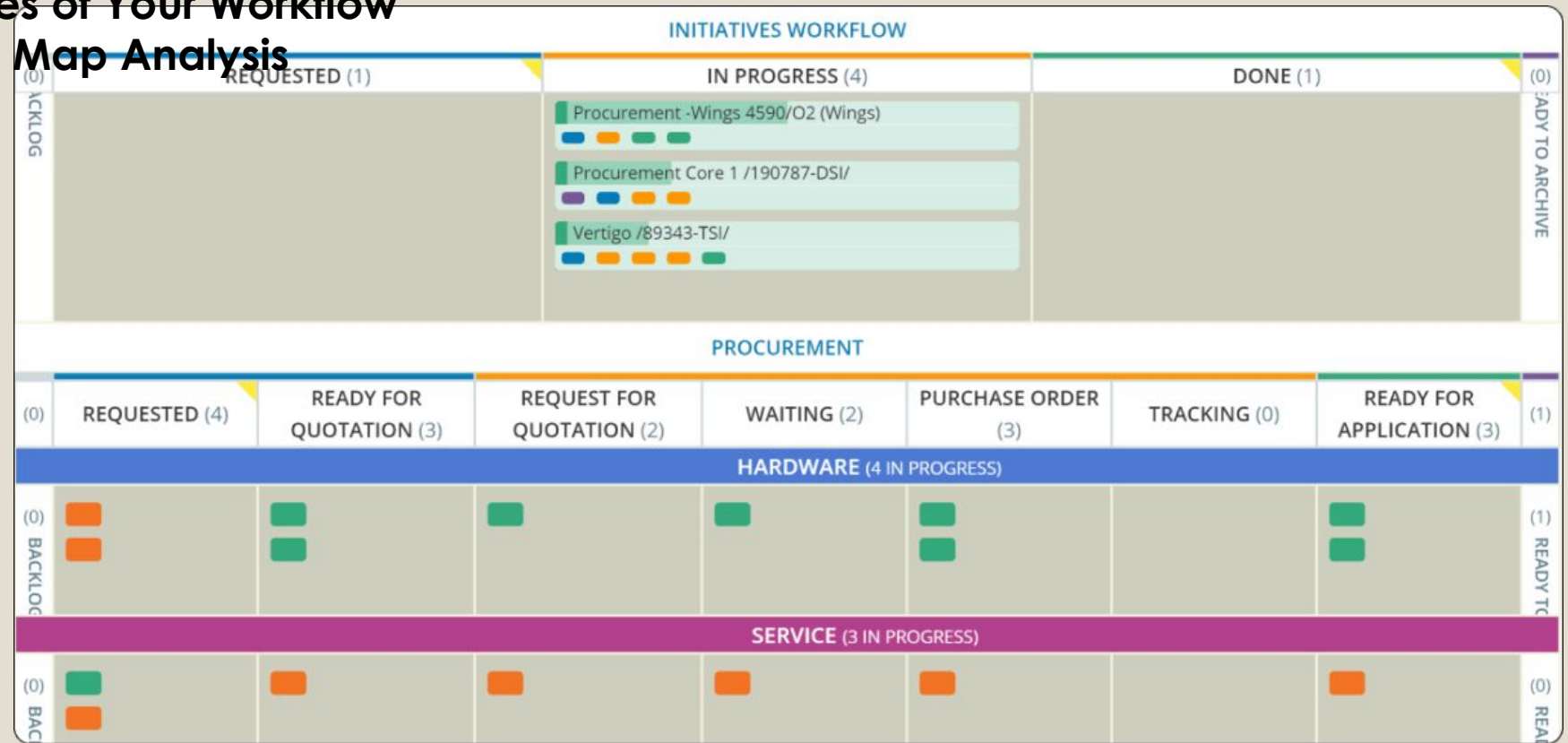
Lean manufacturing	Lean software development
Transportation	Task switching
Inventory	Partially done work
Motion	Hand-offs
Waiting	Delays
Over production	Extra features
Over processing	Extra processing
Defects	Defects
Unused talent	Unused talent

Map the value stream

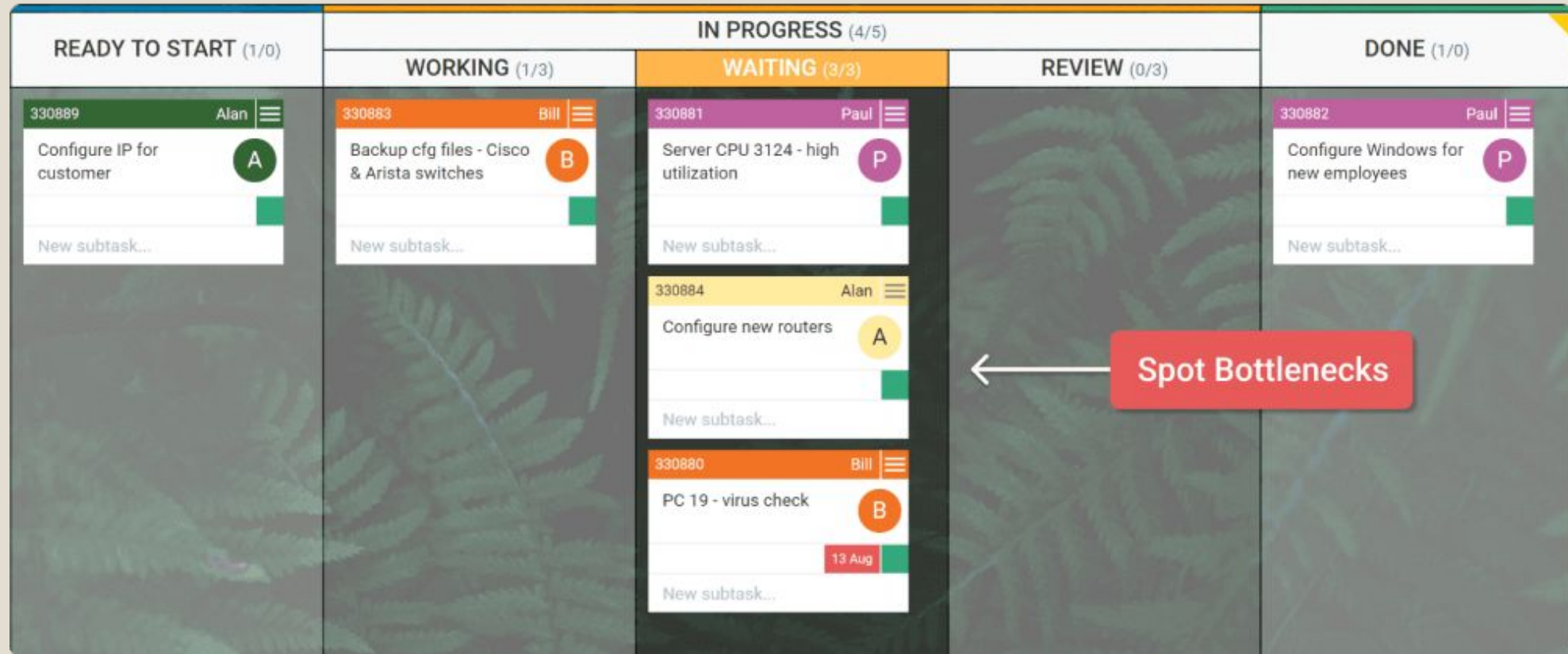
- *"Value stream mapping is a Lean management method that allows you to visualize, analyze and improve all the steps in a product delivery process."*
- Even individual contributors can have a bird's eye overview of how the team's work is progressing.
- Kanban is probably one of the most reliable value stream mapping tools

Value stream mapping

1. Choose a Kanban Board
2. Define VSM Purpose and Expectations
3. Visualize the Key Stages of Your Workflow
4. Perform Value Stream Map Analysis



Create flow



- Having removed all these, your process will be stable and will allow you to use the project resources more efficiently
- Identify steps that do not create value and if you can, eliminate them. If you can't eliminate, improve them. Make the value-creating steps

Create Pull System

- A pull system is a Lean technique for reducing the waste of any production process. Applying a pull system allows you to start new work only when there is customer demand for it. This allows you to reduce overhead and optimize storage costs.
- **a work item has to be in progress only if there is a demand for it.**
- **Example:**

1. Apple is one of the brightest examples of how a pull system can be successful. Have you ever seen these long waiting queues in front of the Apple stores during the iPhone's latest release?
2. Apple always creates a buzz around their new products, and consumers are always ready to buy. They want to pull the product from the stores.
3. Apple doesn't overstock their shops or retail partners. They wait to see if there is a demand for more, and if it increases, they produce more. This way, the company optimizes its resources and achieves high-cost efficiency.

Example -01

- **Truck Manufacturing**

-

Dakota Bodies in South Dakota builds truck bodies for the service and industrial markets. To keep up with market demands, a lean team was established to work with company leaders to define their goals for the lean program. Lean 101 training was provided for supervisors, managers, and key operators. A Kanban system was installed to support just-in-time manufacturing and reduce inventory levels.

Results: Revenues improved by 20 percent and with an overall increase in productivity of 5 percent.

Example 02

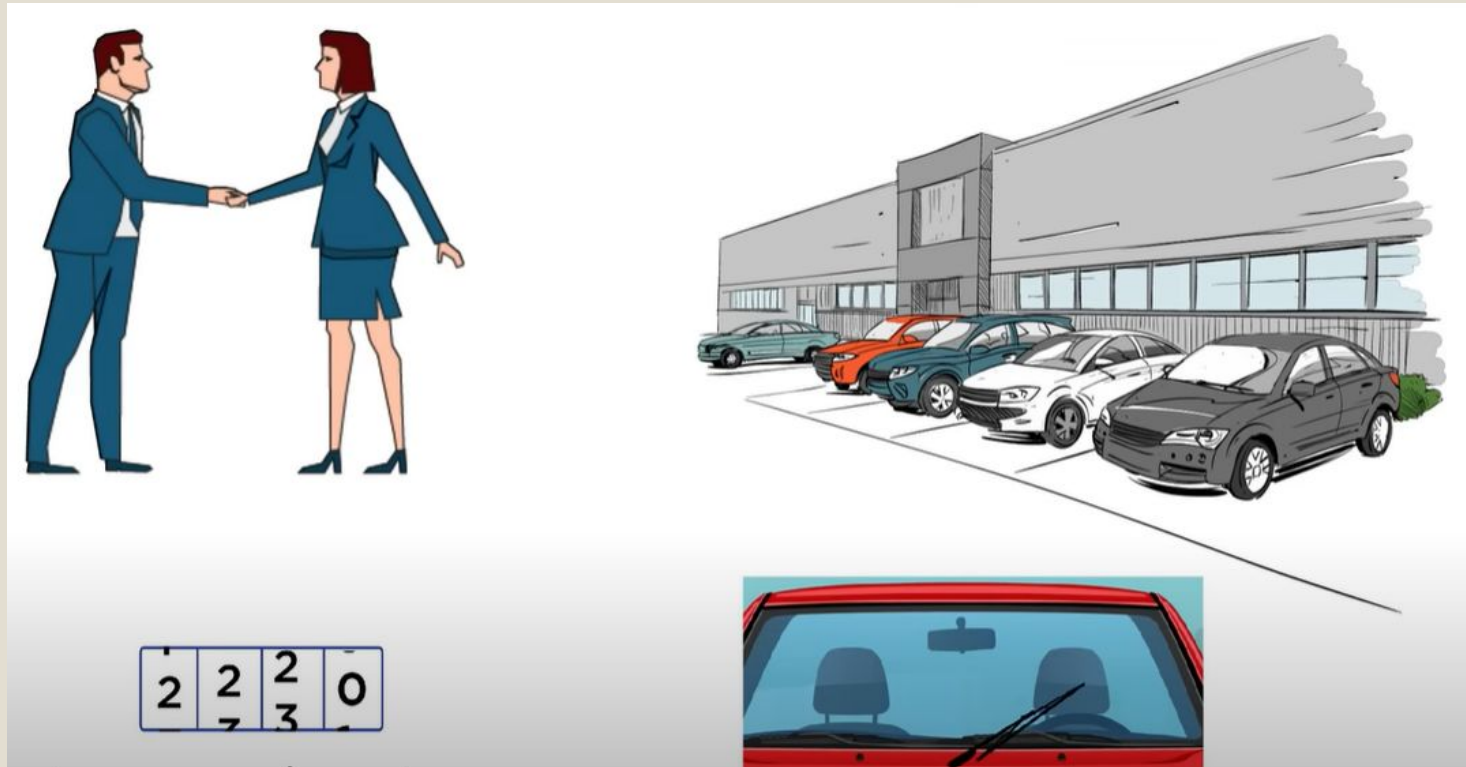
- A printing company was struggling with late orders and long lead times. Inventory piled up in its warehouse, with more than 10 percent of the finished product being thrown away. To improve efficiency, a lean team was assembled to map the value stream and identify waste that could be removed from the workflow. 5S (visual workplace) was also used to maximize workflow before deploying cellular manufacturing systems.

Results: On-time delivery increased to 95 percent with much-shortened lead times; improved inventory management resulted in less waste of finished product

Example

- TOYOTA
- The automobile giant was perhaps the first major company to adopt this lean ideology in their manufacturing processes, initially calling the method the Toyota Production System. Not only have they eliminated waste but they have also mastered the techniques required to minimize faulty products that do not meet customer needs. Toyota works with two primary processes that allow these goals to be reached. The first is a process called **Jidoka**, which translates roughly to “mechanization with the help of humans.” This means that although some aspects of the job are automated, humans are continually checking the quality of the product. There are also programs built into the system that allows the machines to shut themselves down if someone spots a problem. The second part is known as the Just In Time or JIT model. This ensures that the next step of a process is only started once the previous phase is completed. This way, if there is a flaw in the assembly line, no extra and unnecessary work will be completed. This lean manufacturing technique has paved the way for dozens of other companies to follow in their footsteps.

Six Sigma



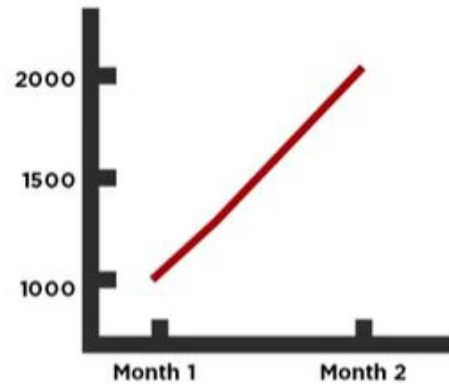
- Six Sigma measures inconsistencies or defects in a process.
- The methodology focuses on perfecting the end product.
- Six Sigma methodologies are used to identify and remove the root cause of defects or reduce the process variation (using statistical or process analysis methods) and also the costs that are associated with it.

Six Sigma

The question you are faced with:



&

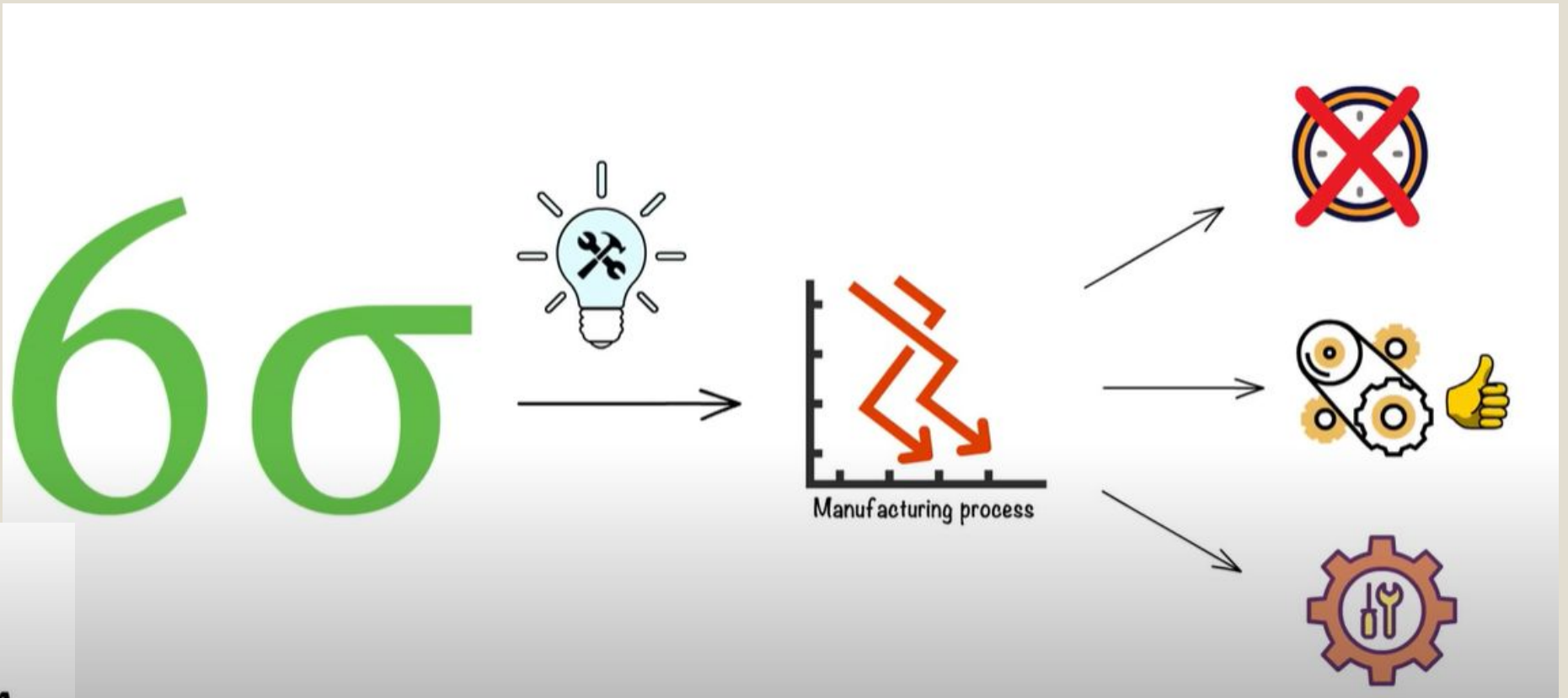


Six Sigma's Approach to Problem Solving

Concept was introduced in 1980



engineer Bill Smith
After that it goes on world wide



Six sigma aims reduce



Time



Defects



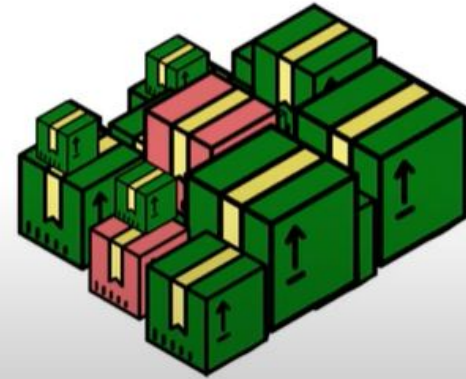
Variability

Six Sigma

Thanks to 6σ

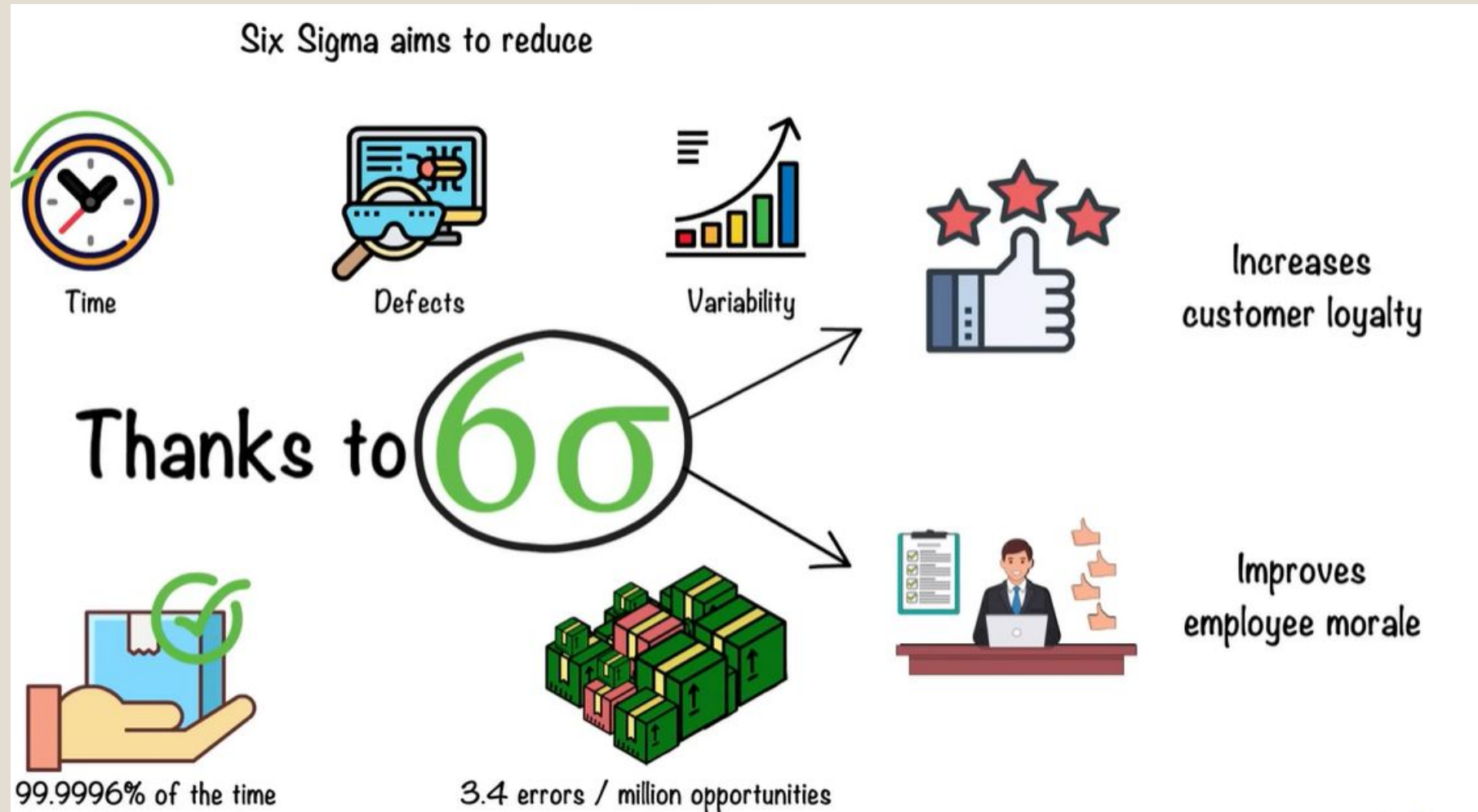


99.9996% of the time

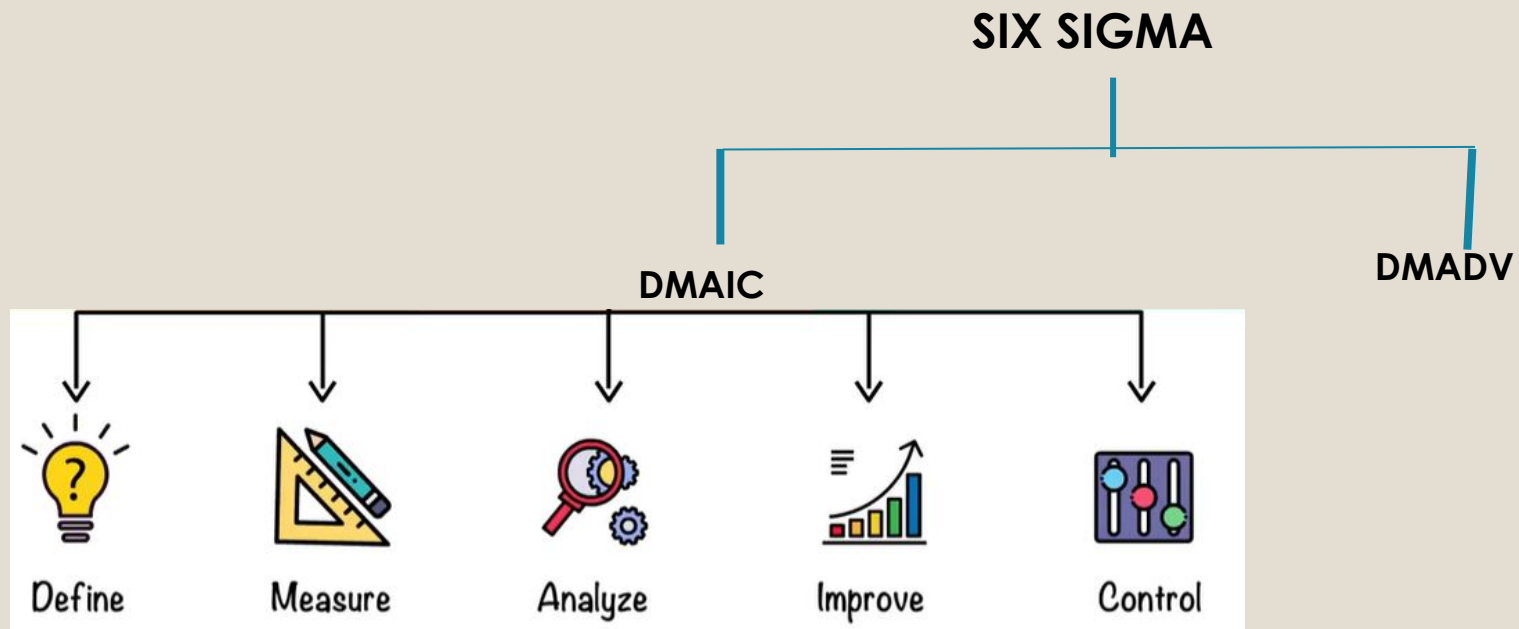


3.4 errors / million opportunities

Six sigma



Six sigma Methodology



DMAIC

Define Phase:



Determine what issues
you are facing



Opportunities for
improvement



Customer
requirements



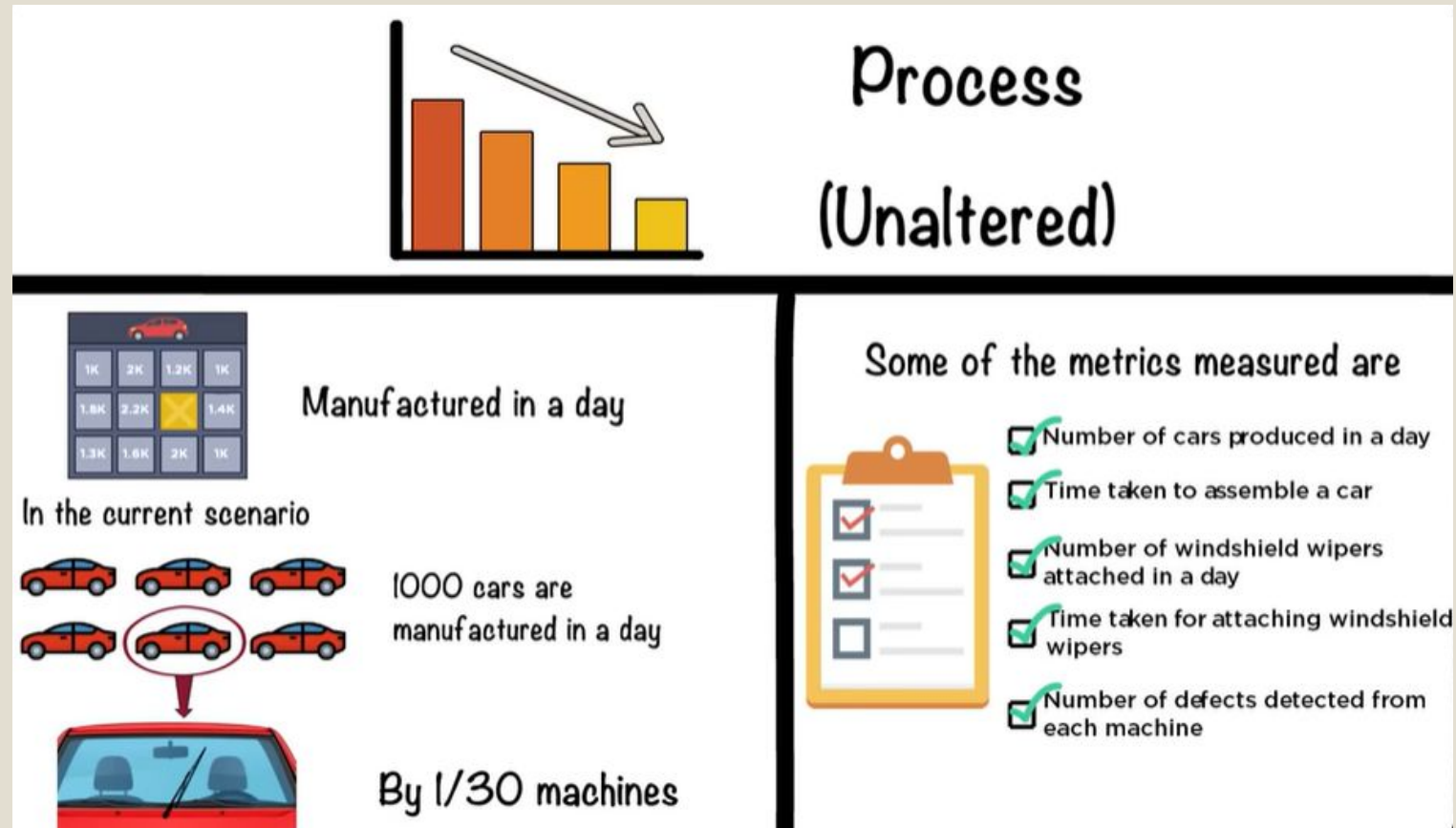
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phase-gate meetings called DMAIC Tollgate Reviews.

DMAIC

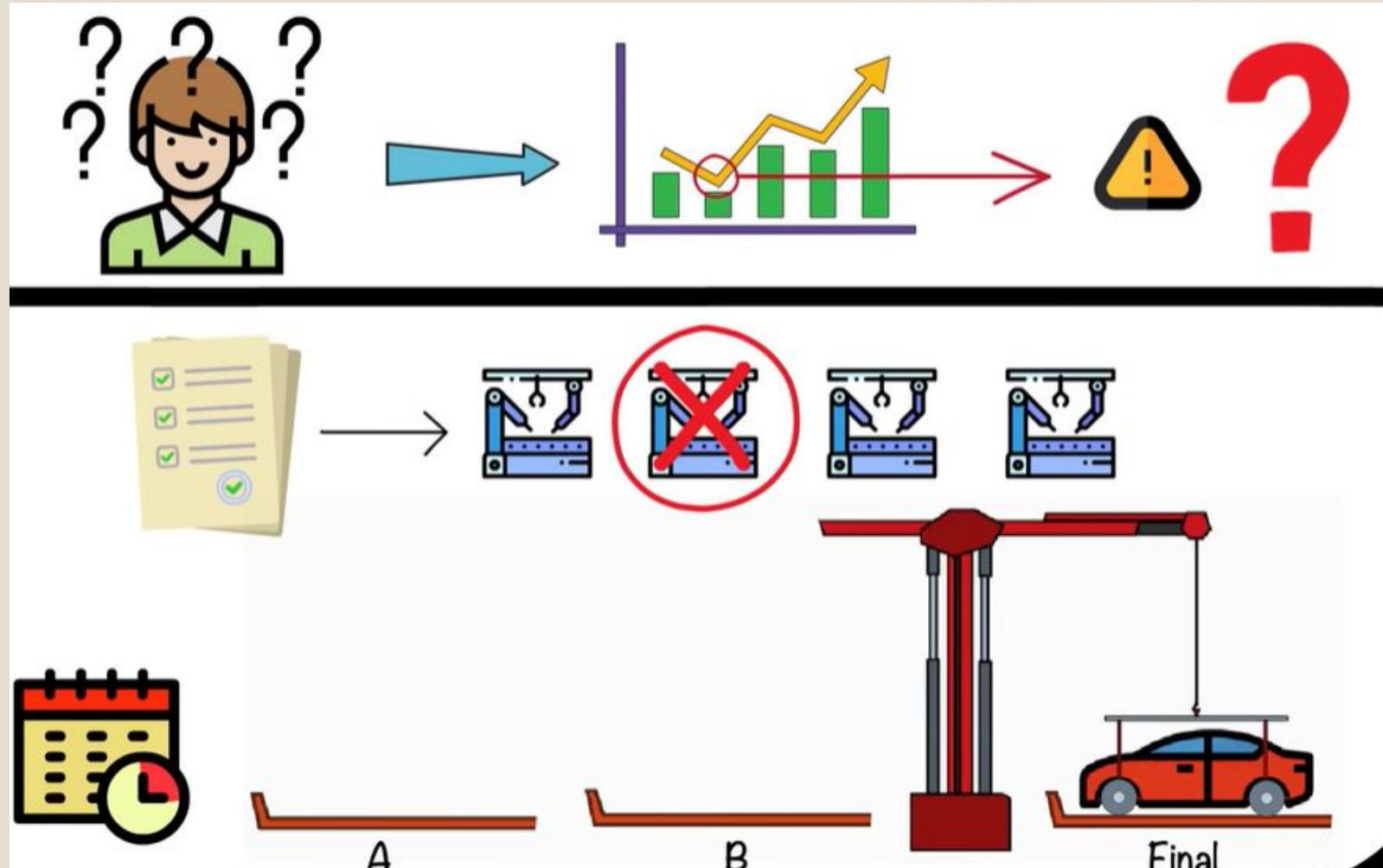
Measure Phase:



phase-gate meetings called DMAIC Tollgate Reviews.

DMAIC

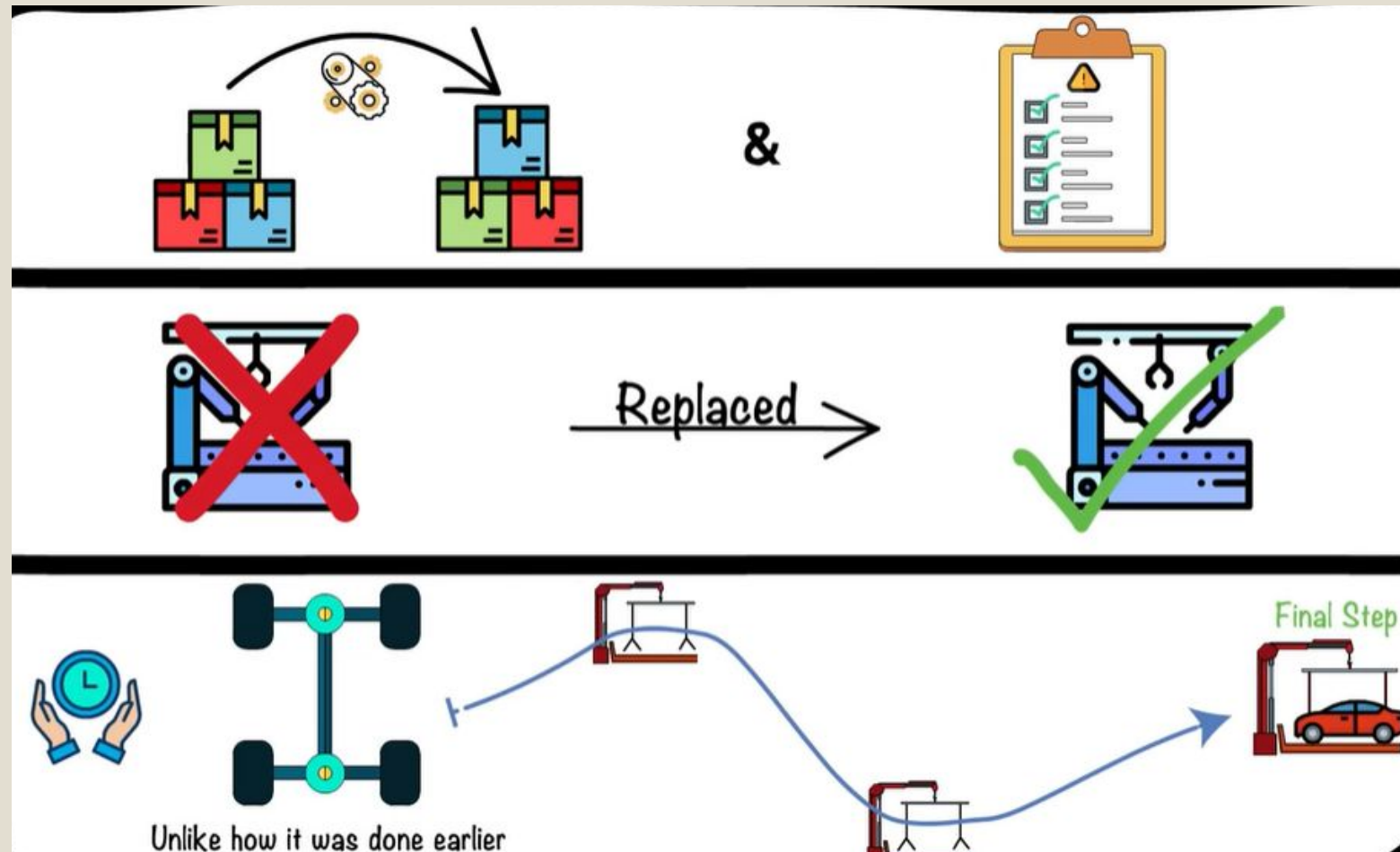
Analyze Phase:



phase-gate meetings called DMAIC Tollgate Reviews.

DMAIC

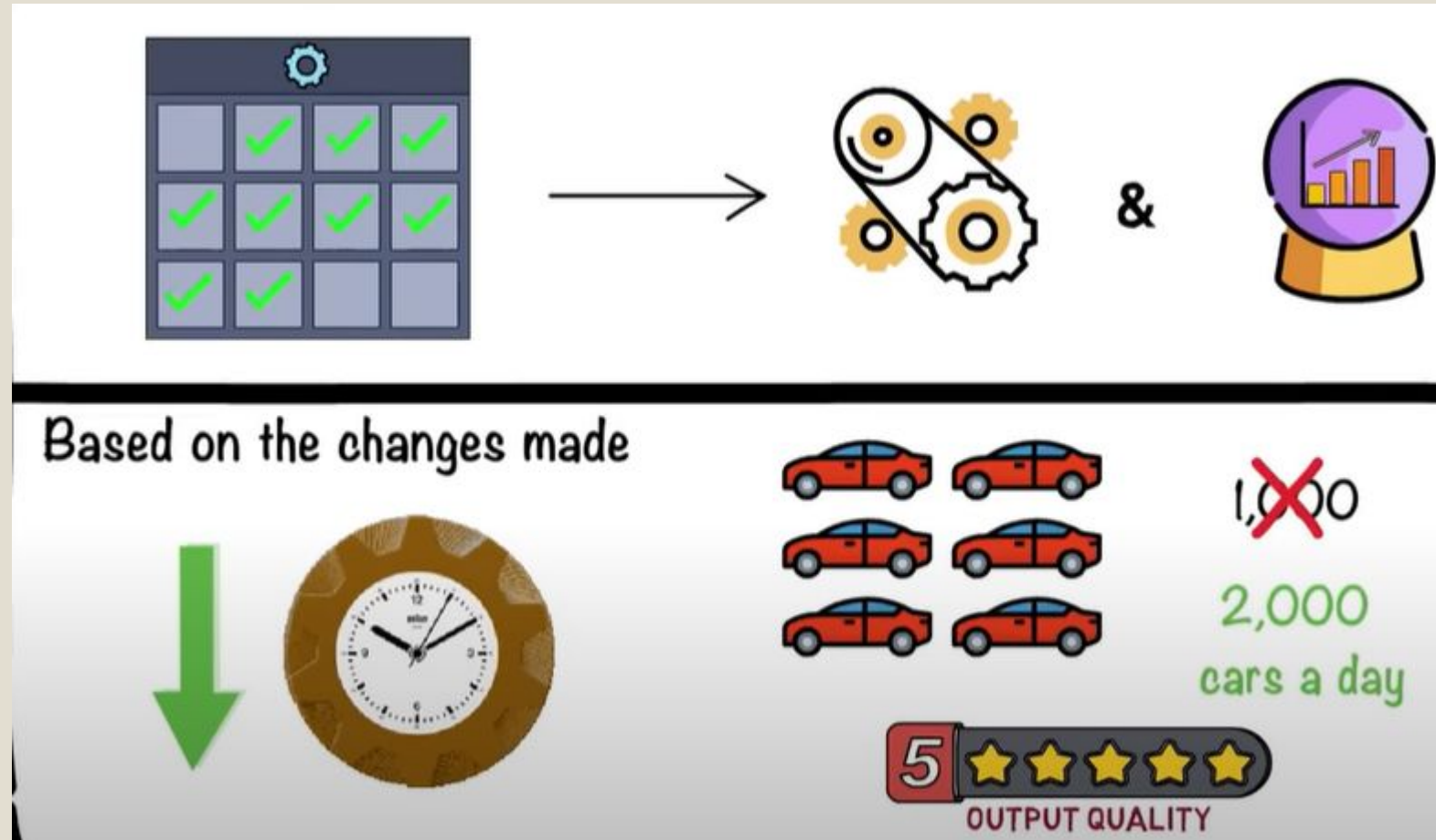
Improve Phase:



phase-gate meetings called DMAIC Tollgate Reviews.

DMAIC

Control Phase:



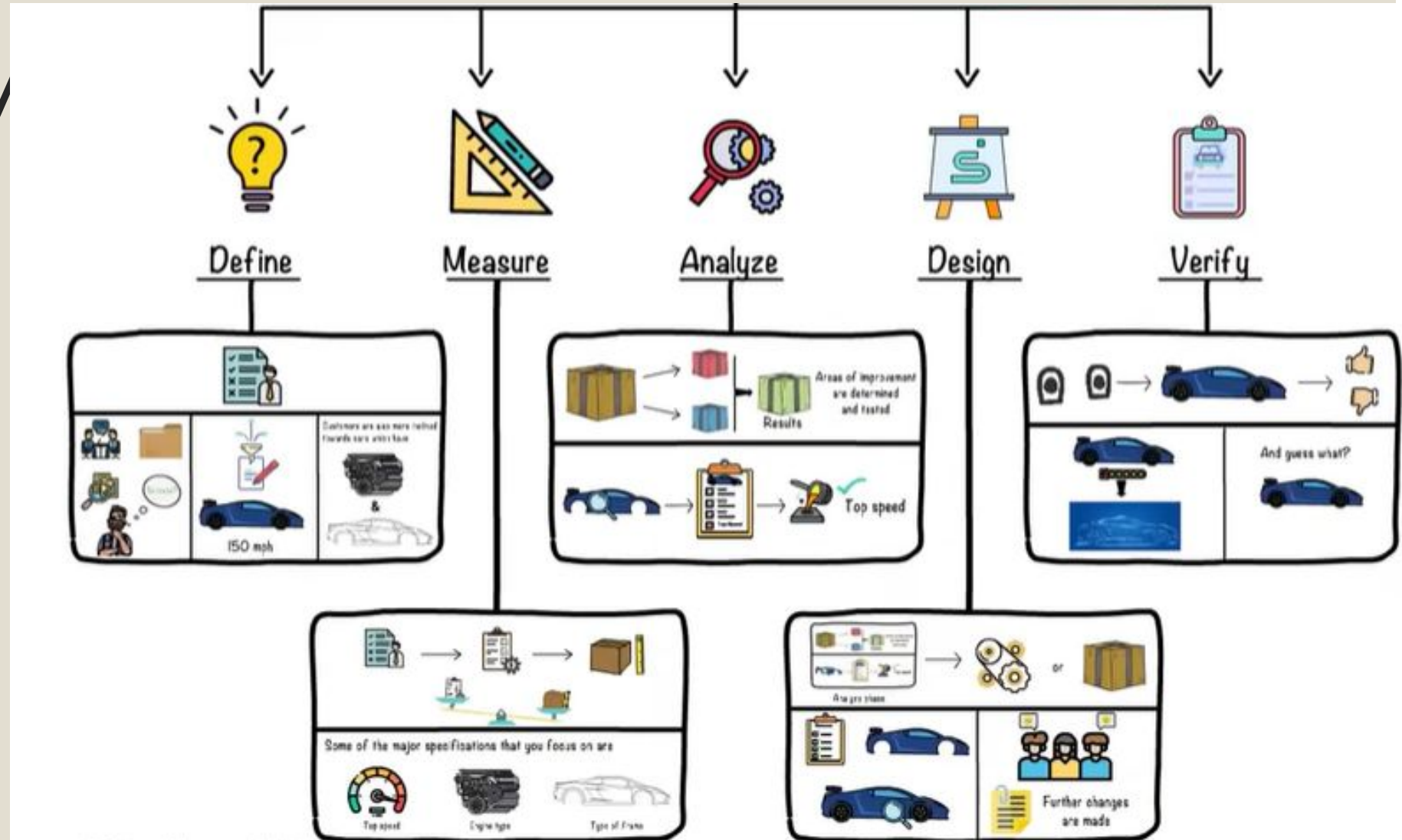
phase-gate meetings called DMAIC Tollgate Reviews.

DMADV

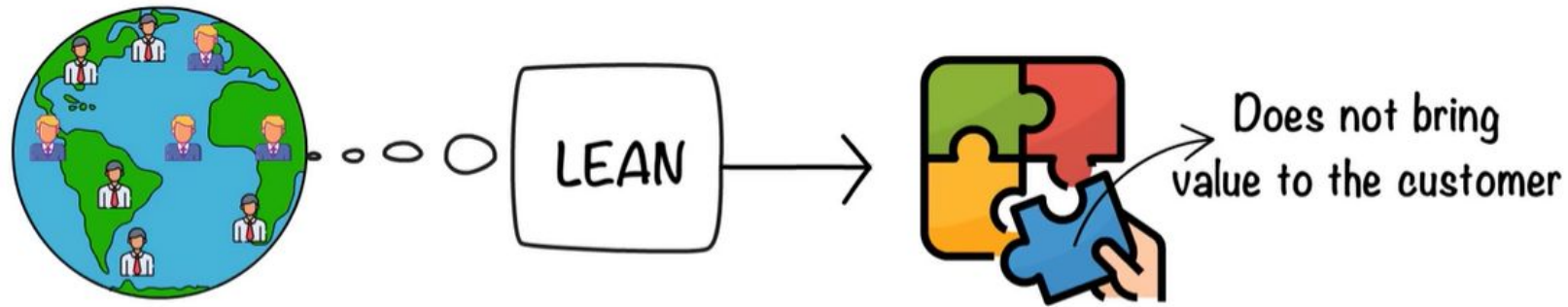
Creating a new product from scratch

It is also called DFSS (Design for six sigma)

Example: Company decide to manufacture a new sports car.



What if (lean + 6 sigma)

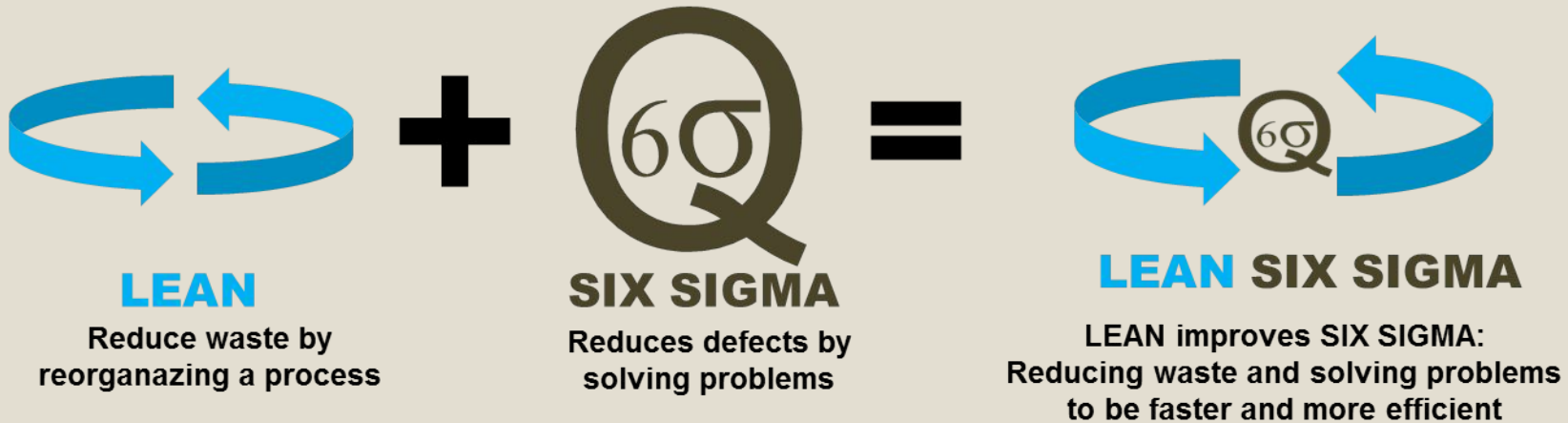


It means doing more with less while doing it better

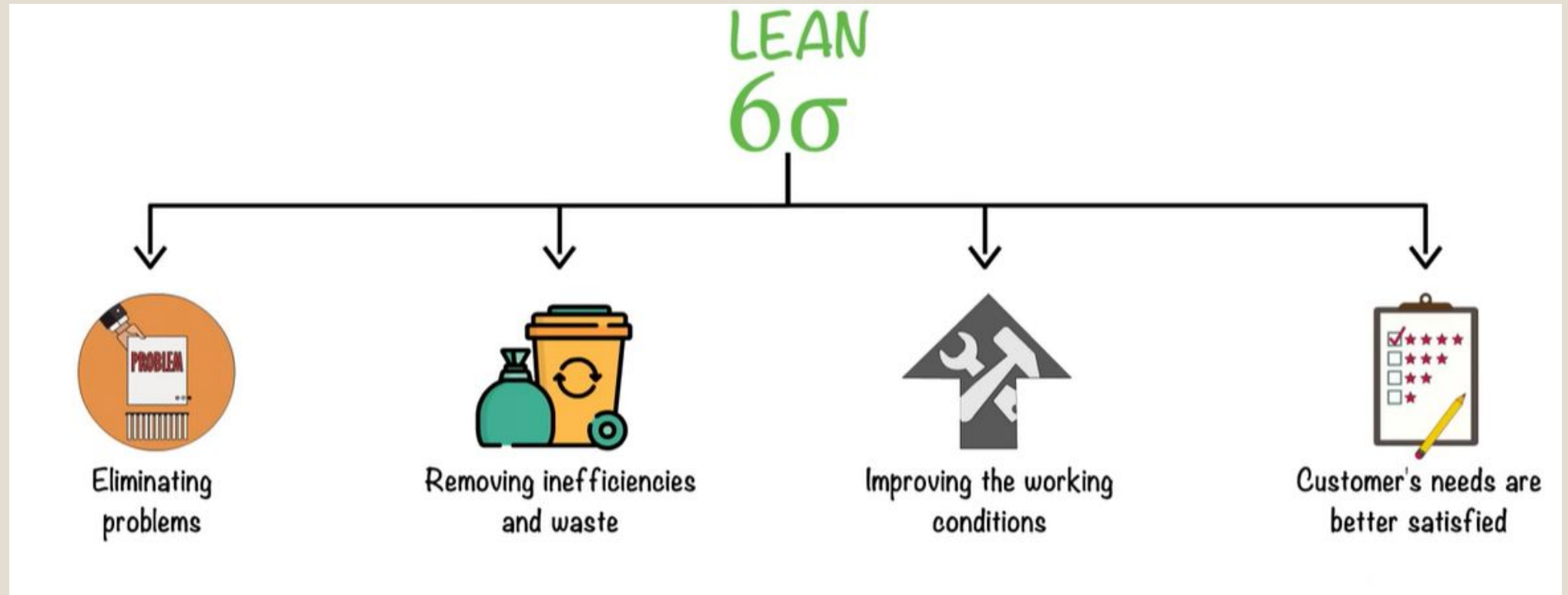


Bob Hartman

What is **LEAN** SIX SIGMA?



Lean Six sigma



Let's Consider

you are the manager of supermarket

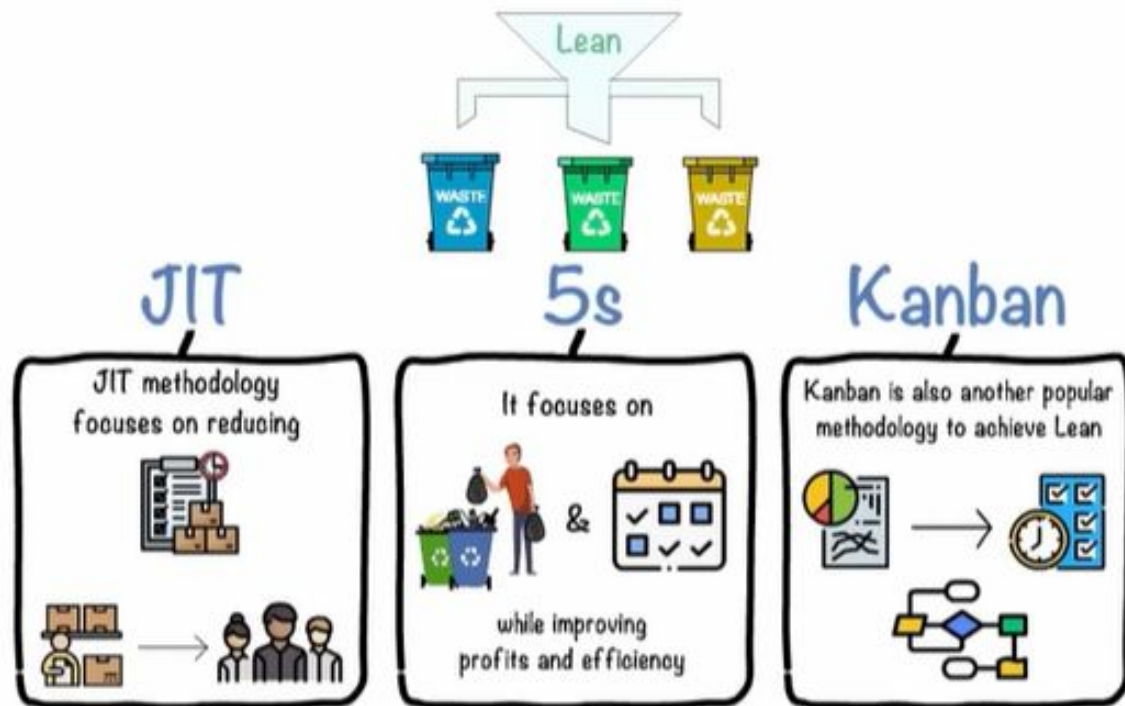
there are two thing thats needs your attention immediately

1. wast of every shop
2. supply chain issues of milk every day morning(resultant delayed delivery and customer dissatisfaction)



Lean Six sigma

1st problem



These methodologies help in optimizing the waste production, and are often used together to maximize results



Lean Six sigma

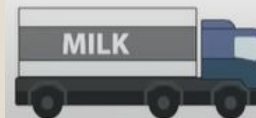
Now the second problem how you can improve the supply efficiency of super store



&



7:30 am → 6:30 am

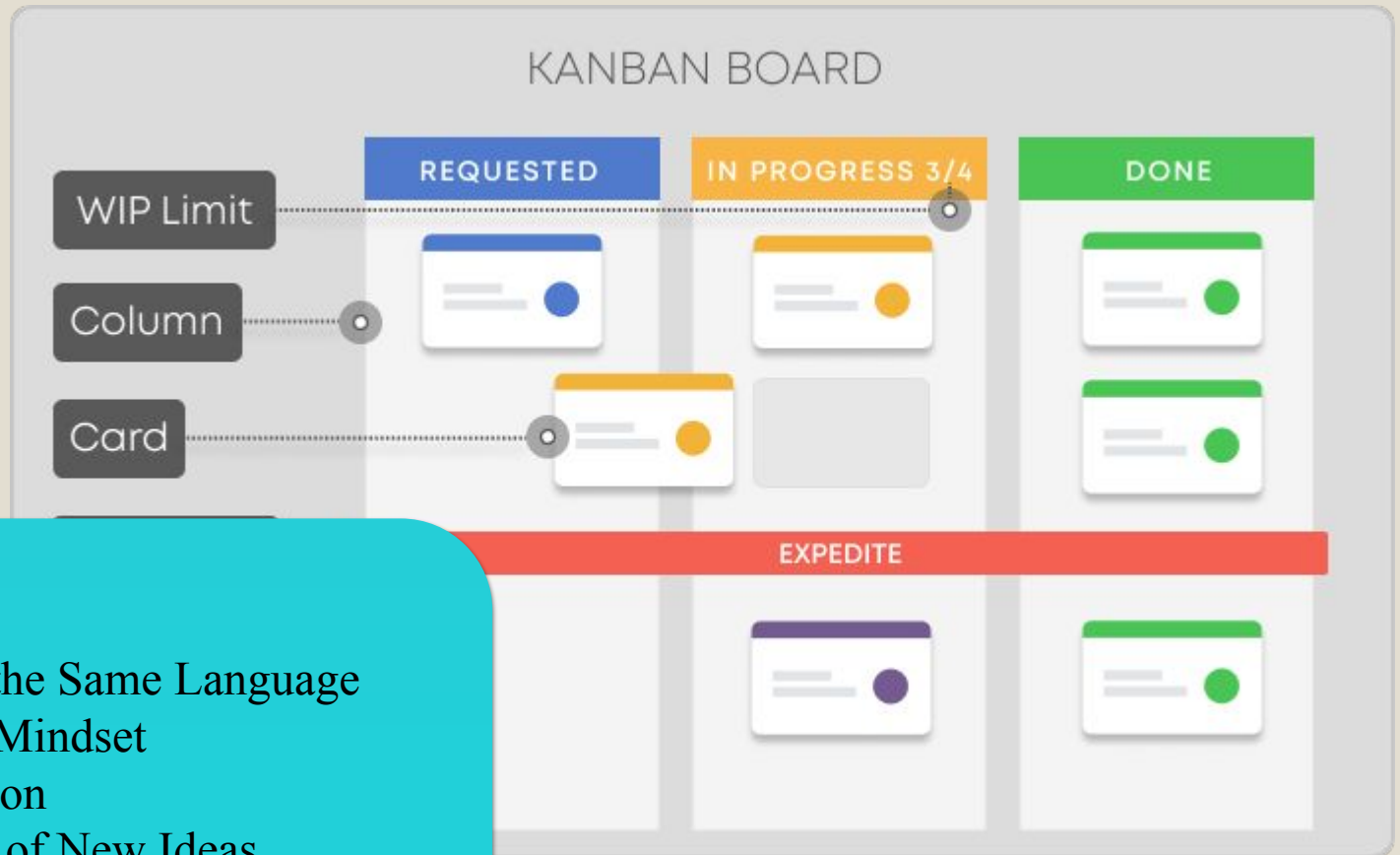


B



What Are the Lean Six Sigma Techniques?

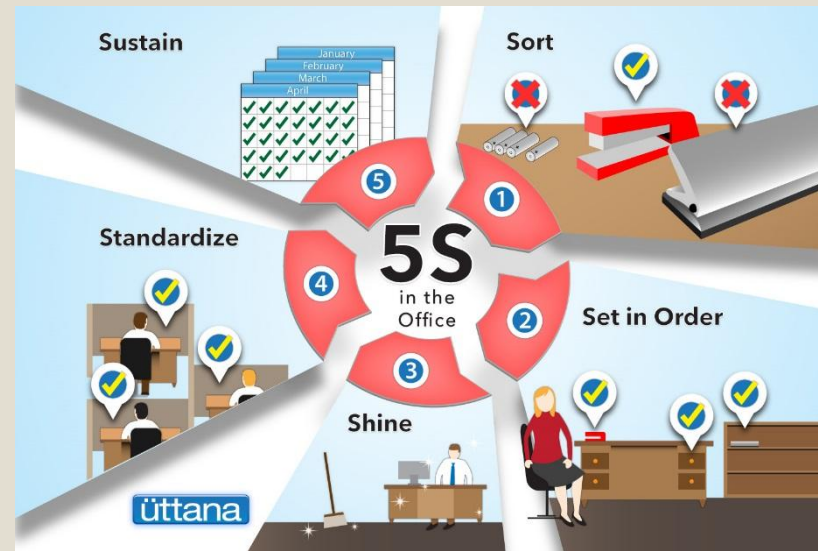
1. Kanban approach
2. Kaizen
3. Value stream mapping
4. 5S model



1. Everyone Speaks the Same Language
2. Creates a Growth Mindset
3. Increases Motivation
4. Better Acceptance of New Ideas

5s of lean

- One of the most popular Lean tool. 5S is a system for organizing spaces so work can be performed efficiently, effectively, and safely. This system focuses on putting everything where it belongs and keeping the workplace clean, which makes it easier for people to do their jobs without wasting time or risking injury.



TQM

- Total Quality Management (TQM) is a methodology that originated in the 1950s, but only becoming popular in the early 1980s.
- As with the Lean methodology, it's main priority is delivering value to the end-customer. The difference is, however, that TQM focuses on the organization as a whole, rather than individual processes. The key idea here is that every department (anything from engineering to marketing) within the company should be optimized with customer values in mind.
- What really makes TQM shine, however, is that it involves the alignment of an entire organization. It's not just the responsibility of the senior management to carry out change – rather, it's a company-wide initiative, all the way down to the lowest level employees.

Total Quality Management Tools

- Total quality management relies on what is known as the Seven Quality Tools, a set of charting and graphing techniques to help identify quality issues. The tools used in TQM are:
 - **Fishbone Diagram / Ishikawa Chart:** Used to visualize cause and effect and identify the root cause.
 - **Check Sheet Template:** Used to collect data in real time.
 - **Control Chart:** Used to check if the process is in a state of control.
 - **Histogram:** Used to estimate probability distribution based on values within a certain range.
 - **Pareto Chart:** Used to evaluate the defects that are frequently occurring and assessing them by category.
 - **Scatter Diagram:** Used to represent two values in a set of data.
 - **Stratification Diagram:** Also called a flow diagram or a run diagram, stratification diagrams are used to sample a group.

Major roadblocks that can influence a company's work processes negatively:

- There are three types of waste in Lean:
 - Muda (wasteful activities)
 - Muri (overburden)
 - Mura (unevenness).

Assignment

